Exact Ranks and Complete Intersections of Four Lucas Sequences

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Abstract

Let $L_0 = 2$, $L_1 = 1$, and $L_{n+2} = L_{n+1} + L_n$. We prove that every positive integer represented as $L_i - 5L_j$ with $i, j \geq 1$ has a unique such representation. We characterize the admissible index pairs, count the values generated by bounded indices exactly, and show that the number of positive values at most x is $\frac{1}{2}(\log_\varphi x)^2 + O(\log x)$. We give an exact count using only three boundary blocks and quadratic formulas for the four nonzero residue subsequences. We completely classify the intersection with A191982 as nine exceptional values and two Lucas-multiple families, derive its exact counting function, classify all five repeated representations of A191982, obtain an exact counting function for A191983, classify the five collisions in A191980, and count its terms exactly. We completely classify the A191980–A191982 intersection into fourteen exceptions and two disjoint Lucas families, derive its exact counting function, and explain the initial zero.

The index region

Write

$$S = \{L_i - 5L_j > 0 : i, j \geq 1\}, \qquad A(x) = \#\{s \in S : s \leq x\}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

The identities

$$L_{n+5} = 5L_{n+1} + 3L_n \quad (n \geq 0), \qquad L_{j+3} - 5L_j = -2L_{j-2} \quad (j \geq 2) \tag{1}$$

follow from the Lucas recurrence (or by checking two initial values and then applying that recurrence).

Theorem 1. *For $i, j \geq 1$, the difference $L_i - 5L_j$ is positive if and only if*

$$(i, j) = (4, 1) \quad \text{or} \quad i \geq 5 \text{ and } 1 \leq j \leq i - 4.$$

For each $i \geq 5$, these $i - 4$ differences are distinct and lie in

$$3L_{i-5} = L_i - 5L_{i-4} \;\leq\; L_i - 5L_j \;\leq\; L_i - 5. \tag{2}$$

Proof. The Lucas numbers increase strictly from index 1 onward. For $j \geq 2$, the second identity in (1) gives $L_{j+3} < 5L_j$, whereas the first gives $L_{j+4} = 5L_j + 3L_{j-1} > 5L_j$. Monotonicity settles all i for such j. For $j = 1$, one checks $L_3 = 4 < 5$ and $L_4 = 7 > 5$. Equation (2) and distinctness within a fixed i follow from monotonicity of L_j, $j \geq 1$. □

Uniqueness and growth

Theorem 2. *Every $s \in S$ has exactly one representation $s = L_i - 5L_j$ with $i, j \geq 1$.*

Proof. The residues $(L_1, L_2, L_3, L_4) \pmod 5$ are $(1,3,4,2)$ and repeat with period four; in particular, they are distinct. Suppose $L_i - 5L_j = L_k - 5L_\ell > 0$. Reduction modulo 5 forces $i \equiv k \pmod 4$. If $i = k$, strict monotonicity of the Lucas numbers from index 1 gives $j = \ell$.

Otherwise, by symmetry take $k \geq i+4$. Theorem 1 gives $i \geq 4$, $k \geq 8$, and $\ell \leq k-4$. Thus

$$L_k - 5L_\ell \geq 3L_{k-5} \geq 3L_{i-1} > L_i > L_i - 5L_j,$$

a contradiction. Here L_n increases for $n \geq 1$, and $3L_{i-1} > L_i = L_{i-1} + L_{i-2}$ for $i \geq 4$. □

Corollary 3. *For every integer $N \geq 4$, precisely*

$$1 + \sum_{i=5}^{N} (i-4) = 1 + \frac{(N-4)(N-3)}{2}$$

positive values have representations with first index $i \leq N$. Furthermore, the elements of S are nonzero modulo 5, and within each fixed nonzero residue class modulo 5, the blocks of values obtained from consecutive permissible first indices occur in strictly increasing order.

Proof. The count follows from Theorems 1 and 2. Nonzero residues follow from the four-term Lucas residue cycle. Two first indices in the same residue class differ by at least four. If $k \geq i+4$, the proof of Theorem 2 shows every value in the k-block exceeds every value in the i-block. □

Theorem 4. *As $x \to \infty$,*

$$A(x) = \frac{1}{2}(\log_\varphi x)^2 + O(\log x).$$

Consequently, if $b_1 < b_2 < \cdots$ enumerates the positive elements of S, then

$$\log_\varphi b_n = \sqrt{2n} + O(1).$$

Proof. Let $m(x)$ be the largest integer $m \geq 1$ with $L_m \leq x$ (for $x \geq 3$). Binet's formula $L_m = \varphi^m + (-\varphi)^{-m}$ implies $m(x) = \log_\varphi x + O(1)$. Every positive difference with $i \leq m(x)$ is at most x, giving

$$A(x) \;\geq\; 1 + \frac{(m(x)-4)(m(x)-3)}{2}$$

for large x. Conversely, a positive difference with $i \geq 5$ is at least $3L_{i-5}$ by (2), so $i \leq m(x/3)+5$. This gives

$$A(x) \;\leq\; 1 + \frac{(m(x/3)+1)(m(x/3)+2)}{2}$$

for large x. Since both $m(x)$ and $m(x/3)$ equal $\log_\varphi x + O(1)$, these bounds prove the asymptotic. Finally, substituting $x = b_n$ and using the uniqueness of all positive terms gives $n = \frac{1}{2}(\log_\varphi b_n)^2 + O(\log b_n)$; solving for $\log_\varphi b_n$ gives the claimed estimate. □

Exact counting and residue-class unranking

For integral y, write $H(y) = \#\{j \geq 1 : L_j \leq y\}$ and $[z]_+ = \max(z, 0)$.

Theorem 5 (Three-block exact counting)**.** *For every integer $x \geq 11$, let m be the unique integer satisfying $L_m \leq x < L_{m+1}$. Then*

$$\boxed{A(x) = 1 + \frac{(m-4)(m-3)}{2} + \sum_{h=1}^{3} \left[m + h - 4 - H\left(\left\lfloor \frac{L_{m+h} - x - 1}{5} \right\rfloor \right) \right]_+ .} \tag{3}$$

For $0 \leq x < 11$, $A(x)$ counts the members of $\{2,3,6,9\}$ not exceeding x.

Proof. All admissible differences with first index $i \le m$ are smaller than $L_i \le x$. For $i \ge m+4$, (2) gives

$$L_i - 5L_j \ge 3L_{i-5} \ge 3L_{m-1} > L_{m+1} > x.$$

Indeed, $3L_{m-1} - L_{m+1} = L_{m-1} - L_{m-2} > 0$ because $m \ge 5$. The only remaining first indices are $i = m+1, m+2, m+3$. In one such block, $L_i - 5L_j \le x$ is equivalent to $L_j > \lfloor (L_i - x - 1)/5 \rfloor$. There are $i-4$ admissible indices j, so removing the ineligible ones gives the positive-part expression in (3). □

Corollary 6 (Exact ranking and membership)**.** *The published OEIS A191981 has $a(1) = 0$. For $n \ge 2$, its n-th term is the least integer x for which $A(x) \ge n-1$. A finite upper bound is L_N, with $N \ge 4$ the least integer such that $1 + (N-4)(N-3)/2 \ge n-1$. For membership, $x > 0$ determines the first index i mod 4 by $x \equiv L_i \bmod 5$. The corresponding index blocks are disjoint and ordered, so a single Lucas lookup for $(L_i - x)/5$ gives the unique witness if one exists.*

Proof. The first assertion follows from uniqueness and (3). At least $n-1$ positive differences have first index at most N and every such difference is smaller than L_N. For the last assertion, the Lucas residues modulo 5 distinguish i mod 4. If $k \ge i+4$, then $\min B_k \ge 3L_{k-5} \ge 3L_{i-1} > L_i > \max B_i$, so blocks with the same first-index residue occur in increasing order. □

Theorem 7 (Quadratic residue-class enumeration)**.** *Let $S_r = \{s \in S : s \equiv r \pmod 5\}$ for $r = 1,2,3,4$, each listed in increasing order. The blocks of S_r have first indices $i_r + 4k$ ($k \ge 0$), where $(i_1, i_2, i_3, i_4) = (5,4,6,7)$. The cumulative numbers of terms through block k are*

$$F_1(k) = (k+1)(2k+1), \quad F_2(k) = 1 + 2k(k+1),$$
$$F_3(k) = 2(k+1)^2, \quad F_4(k) = (k+1)(2k+3).$$

Set $F_r(-1) = 0$. To obtain the n-th term in S_r, choose the least $k \ge 0$ with $F_r(k) \ge n$, put $t = n - F_r(k-1)$ and $i = i_r + 4k$. Then

$$s_{r,n} = L_i - 5L_{i-3-t},$$

except that $s_{2,1} = 2$ (the isolated $i = 4$ block).

Proof. The residue cycle determines i_r. By Corollary 6, successive blocks are ordered; within a block the values increase as j decreases. The block sizes are respectively $1+4k$, $4k$ for $k \ge 1$ (with one exceptional value for $k = 0$), $2+4k$, and $3+4k$. Summing the lengths gives the four quadratics. The t-th element in a nonexceptional block has $j = i - 3 - t$. Inverting a quadratic inequality determines k exactly. □

The complete intersection with A191982

Let $T = \{5L_k - L_\ell > 0 : k, \ell \ge 1\}$ be the positive part of A191982. Unlike S, the set T can have repeated representations: $2 = 5L_1 - L_2 = 5L_3 - L_6$.

Theorem 8 (Complete intersection classification)**.** *With $E = \{2, 6, 9, 12, 13, 14, 24, 32, 61\}$, one has the disjoint union*

$$\boxed{S \cap T = E \,\dot\cup\, \{3L_n : n \ge 6\} \,\dot\cup\, \{8L_n : n \ge 6\}.} \tag{4}$$

In particular, the only common values outside the two Lucas-multiple families are the nine elements of E.

Proof. Suppose $L_i - 5L_j = 5L_k - L_\ell > 0$. The Lucas residues modulo 5 repeat as $(1,3,4,2)$, so $L_i + L_\ell \equiv 0 \pmod 5$ forces $\ell \equiv i+2 \pmod 4$. By Theorem 1, $i \ge 4$ and $j \le i-4$, except that $(i,j) = (4,1)$ is the first permitted pair. Furthermore, $5L_k > L_\ell$ implies $\ell \le k+3$ for every $k \ge 1$. For $k \ge 2$, the least positive difference in its block is $5L_k - L_{k+3} = 2L_{k-2}$.

Assume first $i \geq 16$. The lower bound $L_i - 5L_j \geq 3L_{i-5} > 5L_{i-7}$ excludes $k \leq i-7$. The bound $5L_k - L_\ell \geq 2L_{k-2} \geq 2L_{i-1} > L_i$ excludes $k \geq i+1$. If $k = i$, the largest permissible ℓ is $i+2$ and $5L_i - L_{i+2} > L_i$. If $k = i-2$, the largest permissible ℓ is $i-2$ and $4L_{i-2} > L_i$. Thus $k = i-d$ with $d \in \{1,3,4,5,6\}$. For $d = 1$, only $\ell = i+2$ can give a value below L_i; all smaller permissible ℓ give $5L_{i-1} - L_\ell > L_i$.

Put $R = L_i + L_\ell - 5L_{i-d} = 5L_j$. The following table gives strict open intervals for R; each endpoint is a multiple of a Lucas number.

d	ℓ	location of R
1	$i+2$	$(5L_{i-5}, 5L_{i-4})$
3	$i-2$	$(5L_{i-7}, 5L_{i-6})$
3	$\ell \leq i-6$	$R < 0$
4	$i-2$	$(5L_{i-5}, 5L_{i-4})$
4	$i-6$	$(5L_{i-6}, 5L_{i-5})$
4	$i-10$	$R = 5L_{i-6}$
4	$\ell \leq i-14$	$(5L_{i-7}, 5L_{i-6})$
5	$i-2$	$R > 5L_{i-4}$
5	$\ell \leq i-6$	$(5L_{i-5}, 5L_{i-4})$
6	$i-6$	$R > 5L_{i-4}$
6	$i-10$	$R = 5L_{i-4}$
6	$\ell \leq i-14$	$(5L_{i-5}, 5L_{i-4})$

All table inequalities hold for $i \geq 16$. To verify them, bound each variable ℓ between its indicated upper endpoint and $L_\ell > 0$. The resulting fixed-shift inequalities are differences of Lucas-linear expressions and hence satisfy $u_{i+2} = u_{i+1} + u_i$. In each case the two initial values $i = 16, 17$ are strictly positive, as checked by direct substitution; the recurrence preserves positivity. The two exact-equality rows follow from the same recurrence and two initial identities. Every interval either misses all admissible $5L_j$ or contains no such value, while $R > 5L_{i-4}$ is impossible since $j \leq i-4$. The only surviving cases are

$$(i,j,k,\ell) = (i, i-6, i-4, i-10), \qquad (i, i-4, i-6, i-10) \qquad (i \geq 16).$$

For the remaining $4 \leq i \leq 15$, the finite possibilities $1 \leq k \leq i-1$, $1 \leq \ell \leq k+3$, and $j \leq i-4$ (with $(4,1)$ included) give:

i	4	5	6	7	8	9	10
$S \cap T$ in the i-block	2	6	13	9, 14, 24	12, 32	61	$\varnothing$

For $11 \leq i \leq 15$ the only two values are $3L_{i-5}$ and $8L_{i-5}$. For all $n \geq 6$ the Lucas recurrence gives

$$3L_n = L_{n+5} - 5L_{n+1} = 5L_{n-1} - L_{n-5},$$
$$8L_n = L_{n+5} - 5L_{n-1} = 5L_{n+1} - L_{n-5}.$$

Thus both displayed families belong to $S \cap T$. They are disjoint from one another by the uniqueness theorem for S, and neither family contains one of the nine exceptional values (the smallest values are $3L_6 = 54$ and $8L_6 = 144$). This proves (4). □

Corollary 9 (Exact intersection count)**.** *For $x \geq 0$, let $I(x) = \#\{t \in S \cap T : t \leq x\}$ and $H(y) = \#\{j \geq 1 : L_j \leq y\}$. Then*

$$\boxed{I(x) = \#\{e \in E : e \leq x\} + \big[H(\lfloor x/3 \rfloor) - 5\big]_+ + \big[H(\lfloor x/8 \rfloor) - 5\big]_+.}$$

Consequently

$$I(x) = 2\log_\varphi x + O(1), \qquad \frac{I(x)}{A(x)} \longrightarrow 0 \quad (x \to \infty).$$

Proof. The three sets in (4) are disjoint and the Lucas numbers increase for $n \geq 1$. The count and the asymptotics follow from $H(y) = \log_\varphi y + O(1)$ and Theorem 4. □

All repeated representations in A191982

Theorem 10 (Five and only five collisions)**.** *The map*

$$(k,\ell) \longmapsto 5L_k - L_\ell > 0, \qquad k,\ell \geq 1,$$

is injective except for exactly five pairs of distinct arguments:

k	p	(ℓ, q)	$5L_k - L_\ell = 5L_p - L_q$
1	2	(1,5)	4
1	3	(2,6)	2
2	5	(4,8)	8
2	6	(1,9)	14
3	4	(2,6)	17

In particular, the set of multiply represented positive values in A191982 is precisely $D = \{2,4,8,14,17\}$*; each has multiplicity two.*

Proof. For $k = 1$ the positive pairs have $1 \leq \ell \leq 3$. For every $k \geq 2$, positivity is equivalent to $1 \leq \ell \leq k+3$, and the values in the k-block lie in

$$[2L_{k-2},\ 5L_k - 1], \qquad 5L_k - L_{k+3} = 2L_{k-2}.$$

Suppose $5L_k - L_\ell = 5L_p - L_q > 0$ with $k < p$. Then

$$L_q - L_\ell = 5(L_p - L_k) > 0, \qquad q \equiv \ell \pmod 4. \tag{5}$$

The congruence follows from the four distinct Lucas residues modulo 5. Hence $q \geq \ell + 4$.

If $k \geq 3$ and $p \geq k+4$, then

$$5L_p - L_q \geq 2L_{p-2} \geq 2L_{k+2} > 5L_k - 1,$$

where $2L_{k+2} - 5L_k = L_{k-1} - L_{k-2} > 0$. This exceeds every value in the k-block. For $k = 1$, $p \geq 5$ is excluded by $2L_3 = 8 > 5L_1 - 1 = 4$. For $k = 2$, $p \geq 7$ is excluded by $2L_5 = 22 > 5L_2 - 1 = 14$; the boundary $p = 6$ gives precisely 14.

It remains to consider $p = k + d$ for $d \in \{1,2,3\}$. For $k \geq 4$ the possibilities in (5) reduce as follows.

If $d = 1$, then $5(L_p - L_k) = 5L_{k-1}$, so $L_{k+2} < 5L_{k-1} < L_{k+4}$ and $q \in \{k+3, k+4\}$. For $q = k+3$ one has $L_\ell = 3L_{k-2} > L_{k-1}$, contradicting $\ell \leq q - 4 = k - 1$. For $q = k+4$ one has $L_\ell = L_{k+2} + 3L_{k-2} > L_k$, contradicting $\ell \leq k$.

If $d = 2$, then $5(L_p - L_k) = 5L_{k+1}$. Since $L_{k+4} < 5L_{k+1}$ and $q \leq p+3$, necessarily $q = k+5$. It follows that $L_\ell = 3L_k > L_{k+1}$, but $\ell \leq k+1$.

If $d = 3$, then $5(L_p - L_k) = 10L_{k+1}$. As $L_{k+5} < 10L_{k+1}$ and $q \leq p+3$, one must have $q = k+6$. Thus

$$L_\ell = L_{k+6} - 10L_{k+1} = 5L_k - 2L_{k+1} =: R_k.$$

For $k \geq 3$,

$$L_{k-2} < R_k < L_{k+2}.$$

But (5) forces $\ell \equiv k+2 \pmod 4$ and $\ell \leq k+2$; either $\ell = k+2$ or $\ell \leq k-2$, a contradiction.

Finally, the cases $k = 1,2,3$ can be checked using (5) and $q \leq p+3$: $k = 1$ yields $(p,\ell,q) = (2,1,5), (3,2,6)$; $k = 2$ yields $(5,4,8)$ in addition to the boundary $(6,1,9)$; $k = 3$ yields $(4,2,6)$. Their values are the five displayed in the table. Since the list exhausts all colliding argument pairs, no value has three representations. □

Corollary 11 (Exact counting for A191982 and A191983)**.** *Let $B(x) = \#\{t \in T : t \le x\}$ and let $U(x)$ count the nonnegative terms of A191983 not exceeding x, for integers $x \ge 0$. Put $b_1 = 3$, $b_k = k+3$ for $k \ge 2$, and let $m(y) = \max\{n \ge 1 : L_n \le y\}$, with $m(y) = 0$ if $y < 1$. Define*

$$K(x) = \max\{3, m(\lfloor x/2 \rfloor) + 2\}, \qquad D(x) = \#\{d \in D : d \le x\}.$$

Then both counts are exact:

$$\boxed{B(x) = \sum_{k=1}^{K(x)} [b_k - H(5L_k - x - 1)]_+ - D(x),} \tag{6}$$

$$\boxed{U(x) = 1 + A(x) + B(x) - I(x).} \tag{7}$$

In particular,

$$B(x) = \tfrac{1}{2}(\log_\varphi x)^2 + O(\log x), \qquad U(x) = (\log_\varphi x)^2 + O(\log x).$$

The n-th term in either sequence can be determined exactly by integer binary search on its counting function.

Proof. In the k-block, the positive candidates are $1 \le \ell \le b_k$ and $5L_k - L_\ell \le x$ is equivalent to $L_\ell > 5L_k - x - 1$. Thus the number of eligible pairs in that block is $[b_k - H(5L_k - x - 1)]_+$. For $k \ge 3$ the smallest positive candidate is $2L_{k-2}$, so all $k > K(x)$ contribute nothing. Theorem 10 removes exactly one extra representation for each $d \in D$ not exceeding x. Since the nonnegative union is $\{0\} \cup S \cup T$, (7) follows from inclusion–exclusion and Corollary 9.

Write $t = \log_\varphi x$. All positive pairs with $k \le m((x+1)/5)$ contribute, while none with $k > m(x/2) + 2$ do; both index bounds equal $t + O(1)$. Summing $b_k = k + O(1)$ gives $B(x) = \frac{1}{2}t^2 + O(t)$. The asymptotic for U follows from Theorem 4 and $I(x) = 2t + O(1)$. □

Weighted Lucas sums: A191980

Write $P = \{5L_i + L_j : i, j \ge 1\}$.

Theorem 12 (The five sum collisions)**.** *Exactly five values in P admit two different index-pair representations:*

s	*the two representations of s*
16	$5L_1 + L_5 = 5L_2 + L_1$
23	$5L_1 + L_6 = 5L_3 + L_2$
38	$5L_3 + L_6 = 5L_4 + L_2$
62	$5L_2 + L_8 = 5L_5 + L_4$
91	$5L_2 + L_9 = 5L_6 + L_1$

All other values have precisely one representation; none has three.

Proof. Suppose $5L_i + L_j = 5L_k + L_\ell$ with $(i, j) \ne (k, \ell)$. Since L_n increases strictly for $n \ge 1$, we may take $i < k$ and $j > \ell$. Rearranging gives

$$5L_i - L_\ell = 5L_k - L_j =: t. \tag{8}$$

The four Lucas residues $1, 3, 4, 2$ modulo 5 imply $t \ne 0$. If $t < 0$, then $-t = L_\ell - 5L_i = L_j - 5L_k > 0$ has two distinct representations in S, contradicting Theorem 2. Consequently $t > 0$. Equation (8) is therefore a collision in T, and Theorem 10 restricts it to the five pairs listed there. Conversely, swapping the subtracted Lucas indices in each of those five equalities yields exactly the five sum identities displayed above. Each displayed sum is distinct, and each of the five collisions in T is between precisely two index pairs; hence there are no further collisions and no threefold representations in P. □

Corollary 13 (Exact rank and growth for A191980)**.** *For integer $x \geq 0$ let $C(x) = \#\{p \in P : p \leq x\}$, $H(y) = \#\{j \geq 1 : L_j \leq y\}$, and let $m(y) = \max\{j \geq 1 : L_j \leq y\}$, interpreted as zero for $y < 1$. Then with $D_+ = \{16, 23, 38, 62, 91\}$,*

$$\boxed{C(x) = \sum_{i=1}^{m(\lfloor (x-1)/5 \rfloor)} H(x - 5L_i) - \#\{d \in D_+ : d \leq x\}.} \tag{9}$$

Furthermore,

$$C(x) = (\log_\varphi x)^2 + O(\log x).$$

For $n \geq 1$, the n-th entry of A191980 is the least integer x with $C(x) \geq n$.

Proof. The pairs with $5L_i + L_j \leq x$ have $1 \leq i \leq m(\lfloor (x-1)/5 \rfloor)$, and for each such i there are $H(x - 5L_i)$ choices of j. By Theorem 12, five pairs of representations describe the same values, so one copy must be subtracted for each repeated value not exceeding x. This proves (9). Write $t = \log_\varphi x$. All pairs $i, j \leq m(x/6)$ contribute, and any contributing pair satisfies $i \leq m((x-1)/5)$ and $j \leq m(x-5)$. Both lower and upper counts are $t^2 + O(t)$ by Binet's formula; subtracting at most five terms does not change the asymptotic. Strict ordering gives the rank inversion. □

Theorem 14 (Two infinite sum–difference families)**.** *For every $n \geq 4$, both*

$$u_n = 5L_n + L_{n+7} = 5L_{n+4} - L_{n-3}$$

and

$$v_n = 5L_n + L_{n-3} = 5L_{n+4} - L_{n+7}$$

belong to $P \cap T$. These two families are disjoint, and

$$\#\{z \in P \cap T : z \leq x\} \geq 2\log_\varphi x - O(1) \qquad (x \to \infty).$$

Proof. The identity

$$5(L_{n+4} - L_n) = L_{n+7} + L_{n-3}$$

holds at $n = 4, 5$ by direct substitution. Both sides satisfy the Lucas recurrence in n, so it holds for all $n \geq 4$. Rearrangement gives the two displayed formulas, and $5L_{n+4} - L_{n+7} = 2L_{n+2} > 0$ makes both right-hand sides positive. Both u_n and v_n are strictly increasing as functions of n. If $u_n = v_m$, then the sum would have two distinct representations $(n, n+7)$ and $(m, m-3)$; but $u_n \geq 234 > 91$, contrary to Theorem 12. Thus the families are disjoint. Binet's formula gives $u_n \asymp \varphi^n$ and $v_n \asymp \varphi^n$, so each contributes $\log_\varphi x + O(1)$ different values up to x. □

The complete A191980–A191982 intersection

For $P = \{5L_a + L_b : a, b \geq 1\}$ and $T = \{5L_c - L_d > 0 : c, d \geq 1\}$, put

$$E = \{6, 8, 9, 12, 16, 19, 22, 24, 26, 31, 34, 44, 52, 138\}.$$

Theorem 15 (Complete sum–difference intersection)**.** *The following union is disjoint:*

$$\boxed{P \cap T = E \,\dot\cup\, \{5L_n + L_{n+7} : n \geq 4\} \,\dot\cup\, \{5L_n + L_{n-3} : n \geq 4\}.} \tag{10}$$

Indeed, every equality $5L_a + L_b = 5L_c - L_d > 0$ with $c \geq 8$ has $a = c - 4$ and $\{b, d\} = \{c + 3, c - 7\}$.

Proof. The equality implies $a < c$, since $L_b + L_d > 0$, and

$$L_b + L_d = 5(L_c - L_a), \qquad b \equiv d+2 \pmod 4. \tag{11}$$

The residue condition follows from the four distinct Lucas residues modulo 5. Positivity gives $d \le c+3$; also $L_b < 5L_c < L_{c+4}$ gives $b \le c+3$.

Assume $c \ge 8$, and set $m = \max(b,d)$ and $s = \min(b,d)$. The residue condition forces $m - s \equiv 2 \pmod 4$, so $s \le m-2$. If $m = c+3$, then $s = c+1$ would make $L_m + L_s > 5L_c$, hence $s \le c-3$. If $m = c+2$, then $s \le c$; if $m \le c+1$, then $s \le c-1$. The Lucas recurrence yields

$$L_m + L_s \le L_{c+3} + L_{c-3}, \qquad 5L_c - L_{c+3} - L_{c-3} > 5L_{c-5}.$$

Thus $L_a > L_{c-5}$, hence $a \ge c-4$. Write $h = c-a \in \{1,2,3,4\}$ and $R_h = 5(L_c - L_{c-h}) = L_m + L_s$.

For $h = 1$, the bounds $2L_{c-1} < R_1 < L_{c+2}$ force $m = c$ or $c+1$. When $m = c$, $R_1 > L_c + L_{c-2}$ contradicts $s \le c-2$. When $m = c+1$,

$$L_{c-5} < R_1 - L_{c+1} < L_{c-1}.$$

But then s must belong to $\{c-1, c-5, c-9, \ldots\}$, which is impossible by strict monotonicity. For $h = 2$, the bounds $2L_c < R_2 < L_{c+3}$ force $m = c+1$ or $c+2$. The former is excluded by $R_2 > L_{c+1} + L_{c-1}$; in the latter case

$$L_{c-4} < R_2 - L_{c+2} < L_c,$$

whereas $s \in \{c, c-4, c-8, \ldots\}$. For $h = 3$, $2L_{c+1} < R_3 < L_{c+3}$ forces $m = c+2$, but $R_3 > L_{c+2} + L_c$ contradicts $s \le c$. Finally, for $h = 4$, $R_4 > L_{c+2} + L_c$ forces $m = c+3$; the identity

$$5(L_c - L_{c-4}) = L_{c+3} + L_{c-7}$$

gives $s = c-7$. This gives precisely the two families in (10).

All strict inequalities used above are fixed-shift linear combinations of Lucas numbers. Their differences satisfy $u_{c+2} = u_{c+1} + u_c$; each is positive at $c = 8, 9$, so all hold for $c \ge 8$. The companion verifier records the 18 positive initial comparisons and the exact identity.

For $c \le 7$, the ranges $1 \le a < c$ and $1 \le b, d \le c+3$ are finite. Exhaustion gives the following distinct values, grouped by c:

c	$5L_a + L_b = 5L_c - L_d$ (distinct values)
1	$\varnothing$
2	8, 12
3	9, 16, 19
4	6, 24, 31, 34
5	8, 26, 44, 52
6	$\varnothing$
7	22, 138

These give exactly E. The family with L_{n+7} begins at $234 > \max E$; the other begins with $36, 58, 94, 152, \ldots$, none of which belongs to E. The two families are disjoint by Theorem 12: if one value belonged to both, it would have two different sum representations, yet every value of the first family exceeds 91. Theorem 14 supplies membership of both families in $P \cap T$. □

Corollary 16 (Exact intersection count)**.** *For integers $x \ge 0$, let $J(x) = \#\{z \in P \cap T : z \le x\}$. Set $u_n = 5L_n + L_{n+7}$, $v_n = 5L_n + L_{n-3}$ and*

$$q_u(x) = \max(\{n \ge 4 : u_n \le x\} \cup \{3\}), \quad q_v(x) = \max(\{n \ge 4 : v_n \le x\} \cup \{3\}).$$

Then

$$\boxed{J(x) = \#\{e \in E : e \le x\} + q_u(x) + q_v(x) - 6.} \tag{12}$$

Consequently

$$J(x) = 2\log_\varphi x + O(1), \qquad \frac{J(x)}{C(x)} \longrightarrow 0 \quad (x \to \infty),$$

and the n-th positive common value is the least x with $J(x) \ge n$.

Proof. Both families are strictly increasing and disjoint, so (12) follows from Theorem 15. Binet's formula gives $u_n \asymp \varphi^n$ and $v_n \asymp \varphi^n$; therefore $q_u(x) = \log_\varphi x + O(1)$ and similarly for q_v. The density statement follows from $C(x) = (\log_\varphi x)^2 + O(\log x)$. □

Remark 17 (The initial zero)**.** *A191981 begins with* 0*, but* 5 *never divides a Lucas number: $L_i \equiv 1,3,4,2 \pmod 5$ for $i \ge 1$. Hence $L_i - 5L_j = 0$ has no solution. The Mathematica program linked from A191980 replaces nonpositive differences by zero before taking their union. Thus the published sequence is $\{0\} \cup S$, in increasing order; its zero is a program-generated sentinel, not a Lucas difference. In particular, the number of published terms at most x is $A(x) + 1$ for $x \ge 0$.*

Reference. C. Kimberling, *OEIS A191981: Ordered sequence of nonnegative differences $f - 5g$, where f and g are Lucas numbers*, https://oeis.org/A191981 (accessed October 2026). The related sequences are https://oeis.org/A191980, https://oeis.org/A191982, and https://oeis.org/A191983; the Mathematica code appears under https://oeis.org/A191980.